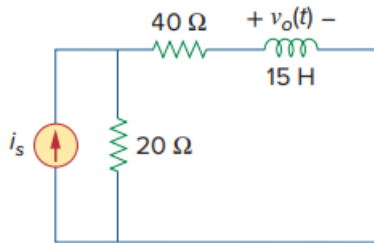


Problem

Determine the step response $v_o(t)$ to $i_s = 6u(t)$ A in the circuit of Fig. 7.124.

Fig. 7.124:



Step-by-step solution

Step 1 of 5

Refer to figure 7.124 in the textbook.

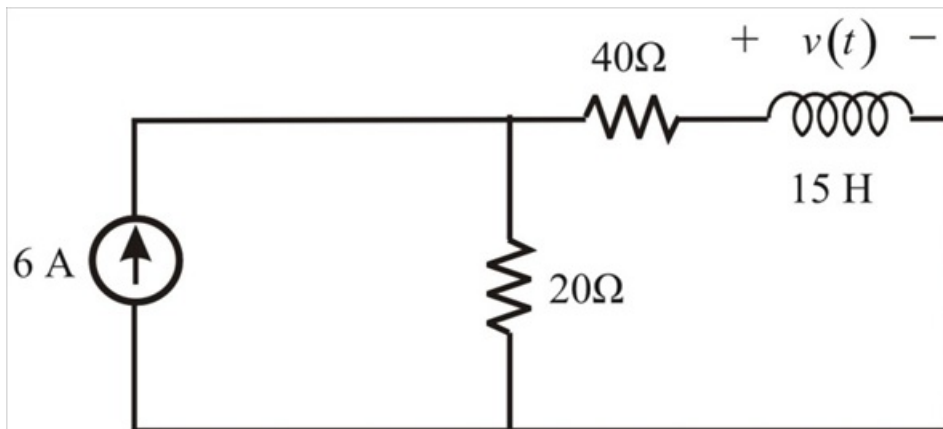
Consider the step input.

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}$$

Consider that the circuit is source free network for $t < 0$. Therefore, the initial current flowing through the inductor is,

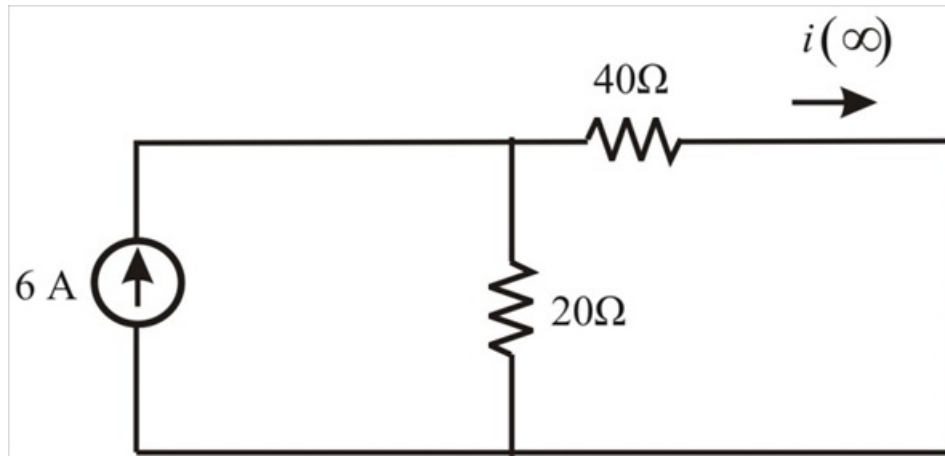
$$i(0) = 0\text{ A}$$

Draw the circuit for $t > 0$ as shown in the following figure:



Step 2 of 5

Consider that the circuit is in that position for a long time the circuit reached steady state and inductor acts like an short circuit.



Step 3 of 5

Calculate the steady state value of inductor current.

Apply the current division rule.

$$i(\infty) = 6 \times \frac{20}{20 + 40}$$

$$i(\infty) = 2 \text{ A}$$

Therefore, the inductor current is,

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}}$$

$$= 2 + [0 - 2]e^{-\frac{t}{\tau}}$$

$$i(t) = 2 \left(1 - e^{-\frac{t}{\tau}} \right) \text{ A}$$

Here, τ is time constant.

Step 4 of 5

Calculate the time constant.

$$\tau = \frac{L}{R_{eq}}$$

Calculate the equivalent resistance seen from the inductor for $t > 0$.

$$R_{eq} = 40 + 20$$

$$R_{eq} = 60 \Omega$$

$$\tau = \frac{15}{60}$$

$$\tau = \frac{1}{4} \text{ sec}$$

Therefore, the inductor current for $t > 0$ is,

$$i(t) = 2 \left(1 - e^{-4t} \right) \text{ A}$$

Step 5 of 5

Calculate the voltage across the inductor.

$$\begin{aligned}v_o(t) &= L \frac{di(t)}{dt} \\&= 15 \frac{d}{dt} (2(1 - e^{-4t})) \\&= -15(2)(-4)e^{-4t} \\v_o(t) &= 120e^{-4t} \text{ V}\end{aligned}$$

Therefore, the value of voltage $v_o(t)$ for $t > 0$ is $\boxed{120e^{-4t} \text{ V}}$.